

Final Report

Aidan Frederick, Ben Walter, Kyle Maher

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Packages

```
library(fpp3) # Time Series Plots and Forecasting
library(corrplot) # Correlation Plot
library(forecast) # Box-Cox Transformation
library(scales) # Log Scale Tick Lables
set.seed(280) # Reproducibility
```

Setup

Read Data

```
data <- readRDS("../data/loaded/delhi.rds")
```

Filter to Delhi

```
delhi_city <- data %>%
  filter(file_name == "DL008.csv")
```

Determine Parameter Counts

```
observation_counts <- delhi_city %>%
  summarise(across(everything(), ~sum(!is.na(.)))) %>%
  pivot_longer(cols = everything()) %>%
  arrange(desc(value)) %>%
  mutate(percent_available = (value / nrow(delhi_city)) * 100)
```

```
observation_counts
```

```
# A tibble: 60 x 3
```

name	value	percent_available
<chr>	<int>	<dbl>

1	From Date	114635	100
2	To Date	114635	100
3	file_name	114635	100
4	CO (mg/m3)	48443	42.3
5	Benzene (ug/m3)	48136	42.0
6	NOx (ppb)	48007	41.9
7	Ozone (ppb)	47576	41.5
8	Toluene (ug/m3)	46987	41.0
9	PM2.5 (ug/m3)	46610	40.7
10	Xylene (ug/m3)	46564	40.6

i 50 more rows

Drop Parameters Available For Less than 2% of Hours, Rename Columns, & Filter Date Range

```
rare_parameters <- observation_counts %>%
  filter(percent_available < 2) %>%
  pull(name)

df <- delhi_city %>%
  select(-all_of(rare_parameters), -file_name, -`From Date`, -`Ozone (ppb)`) %>%
  rename(
    "datetime" = "To Date",
    "PM2.5" = "PM2.5 (ug/m3)",
    "PM10" = "PM10 (ug/m3)",
    "NO" = "NO (ug/m3)",
    "NO2" = "NO2 (ug/m3)",
    "NOx" = "NOx (ppb)",
    "CO" = "CO (mg/m3)",
    "Benzene" = "Benzene (ug/m3)",
    "Toluene" = "Toluene (ug/m3)",
    "Xylene" = "Xylene (ug/m3)"
  ) %>%
  as_tsibble(index = datetime) %>%
  filter(datetime >= as_datetime("2017-08-31 01:00:00"))
```

Show Hourly Data Availability

```
df %>%
  as_tibble() %>%
  summarise(across(everything(), ~sum(!is.na(.)))) %>%
  pivot_longer(cols = everything()) %>%
  arrange(desc(value)) %>%
  mutate(percent_available = (value / nrow(df)) * 100)
```

```
# A tibble: 10 x 3
  name      value percent_available
  <chr>    <int>          <dbl>
1 datetime 48936          100
2 CO       44884           91.7
3 Toluene  44635           91.2
4 Benzene  44626           91.2
5 NOx      44583           91.1
6 Xylene   44212           90.3
7 PM2.5    43227           88.3
8 PM10     42937           87.7
9 NO2      42161           86.2
10 NO      41407           84.6
```

Aggregate to Daily Series

```
daily_df <- df %>%
  index_by(date = as.Date(datetime)) %>%
  summarise(
    PM2.5 = mean(PM2.5, na.rm = TRUE),
    PM10 = mean(PM10, na.rm = TRUE),
    NO = mean(NO, na.rm = TRUE),
    NO2 = mean(NO2, na.rm = TRUE),
    NOx = mean(NOx, na.rm = TRUE),
    CO = mean(CO, na.rm = TRUE),
    Benzene = mean(Benzene, na.rm = TRUE),
    Toluene = mean(Toluene, na.rm = TRUE),
    Xylene = mean(Xylene, na.rm = TRUE)
  )
```

Show Daily Availability

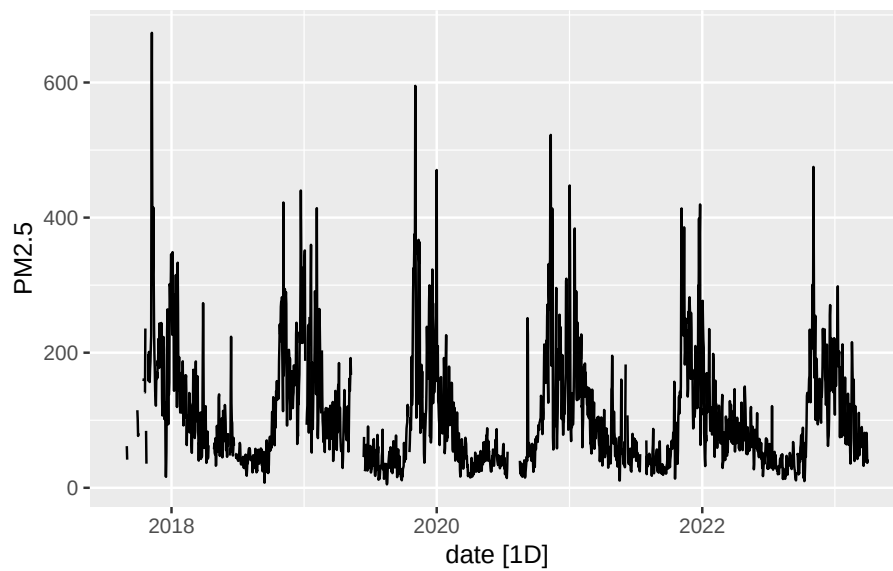
```
daily_df %>%
  as_tibble() %>%
  summarise(across(everything(), ~sum(!is.na(.)))) %>%
  pivot_longer(cols = everything()) %>%
  arrange(desc(value)) %>%
  mutate(percent_available = (value / nrow(daily_df)) * 100)
```

```
# A tibble: 10 x 3
  name      value percent_available
  <chr>    <int>          <dbl>
1 date      2040          100
2 CO        1927           94.5
3 NOx       1925           94.4
```

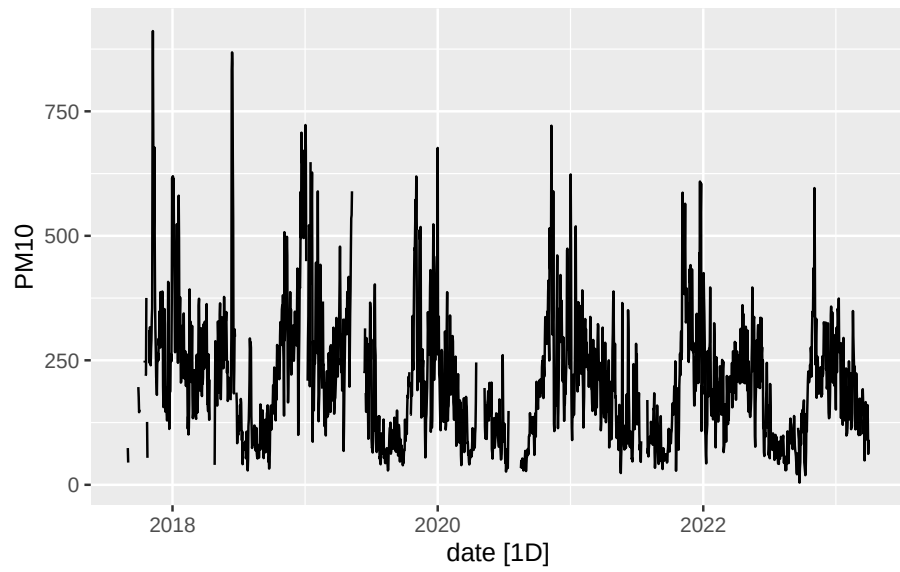
4	Benzene	1911	93.7
5	Toluene	1911	93.7
6	Xylene	1892	92.7
7	PM2.5	1884	92.4
8	PM10	1867	91.5
9	NO2	1841	90.2
10	NO	1818	89.1

Plot Each Parameter

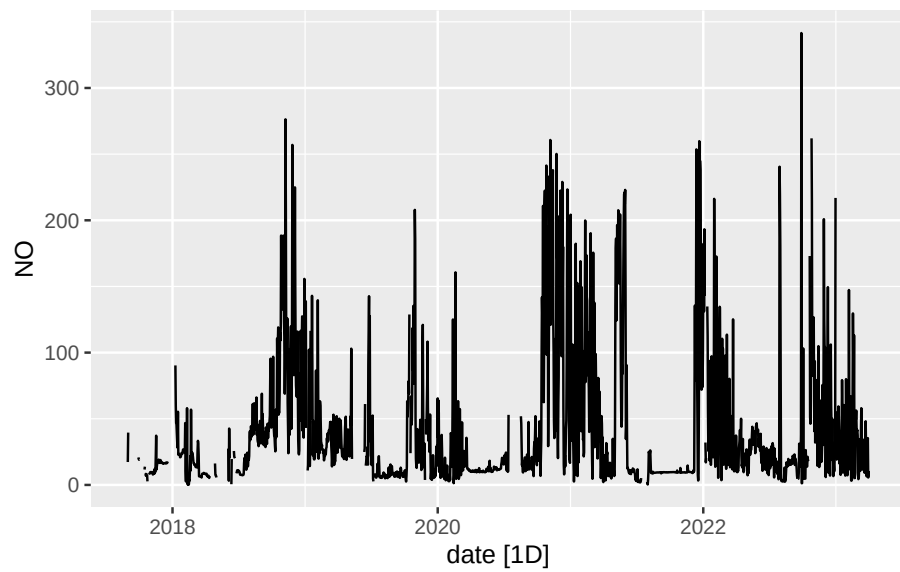
```
daily_df %>%
  autoplot(PM2.5)
```



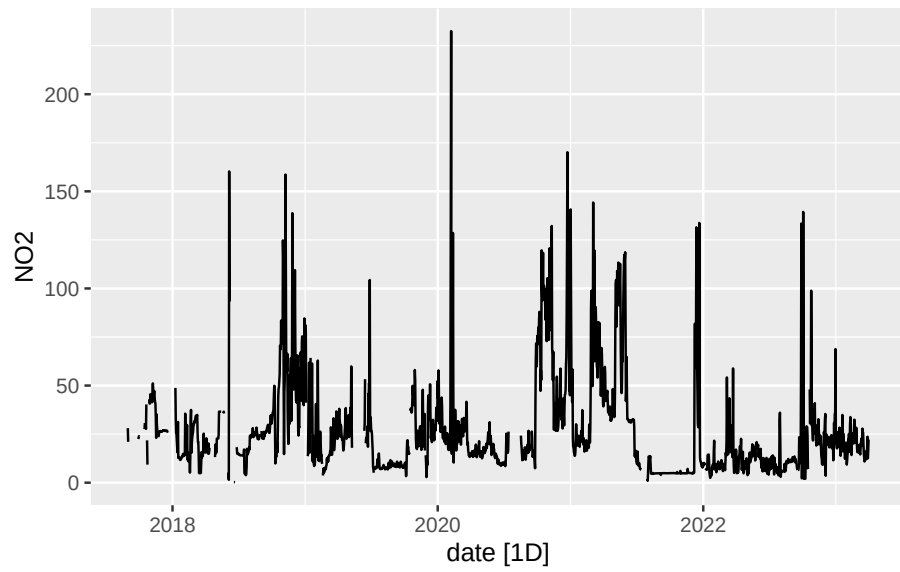
```
daily_df %>%
  autoplot(PM10)
```



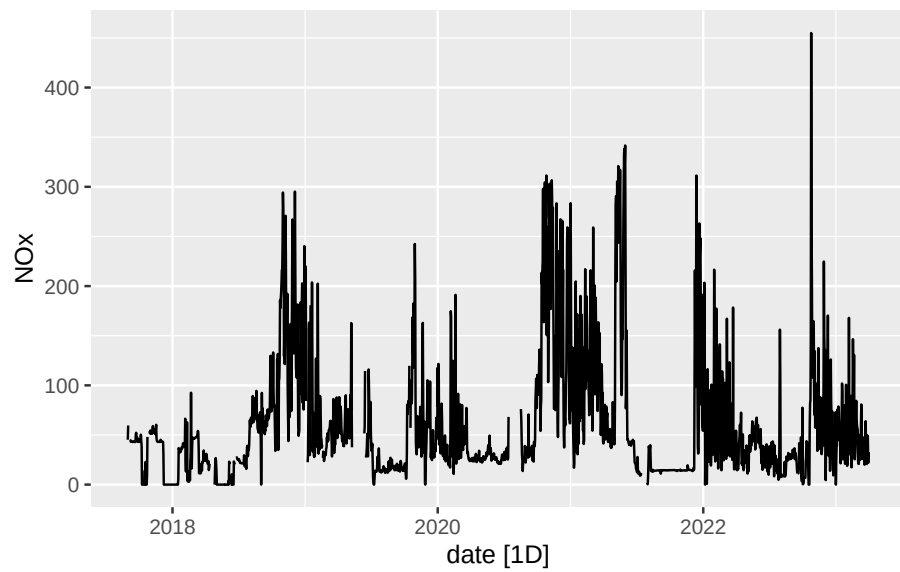
```
daily_df %>%  
  autoplot(NO)
```



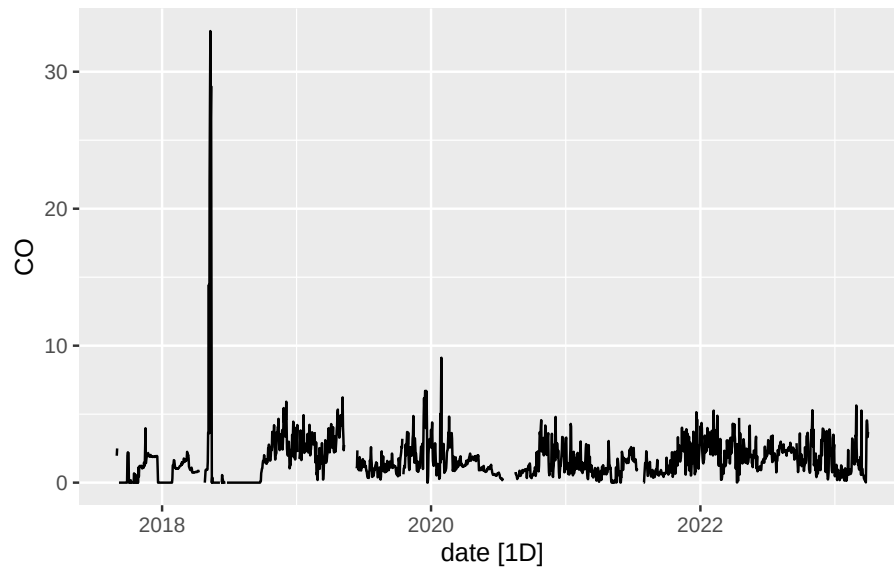
```
daily_df %>%  
  autoplot(NO2)
```



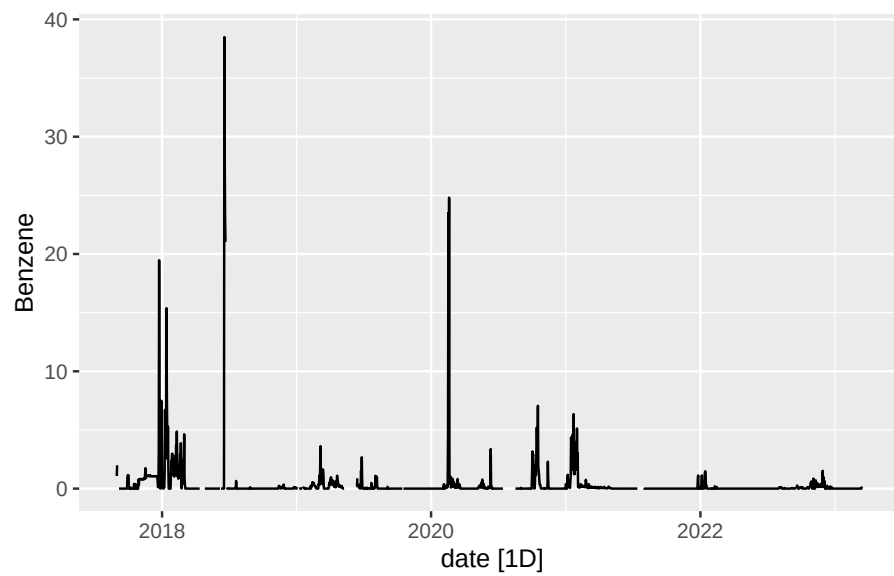
```
daily_df %>%  
  autoplot(NOx)
```



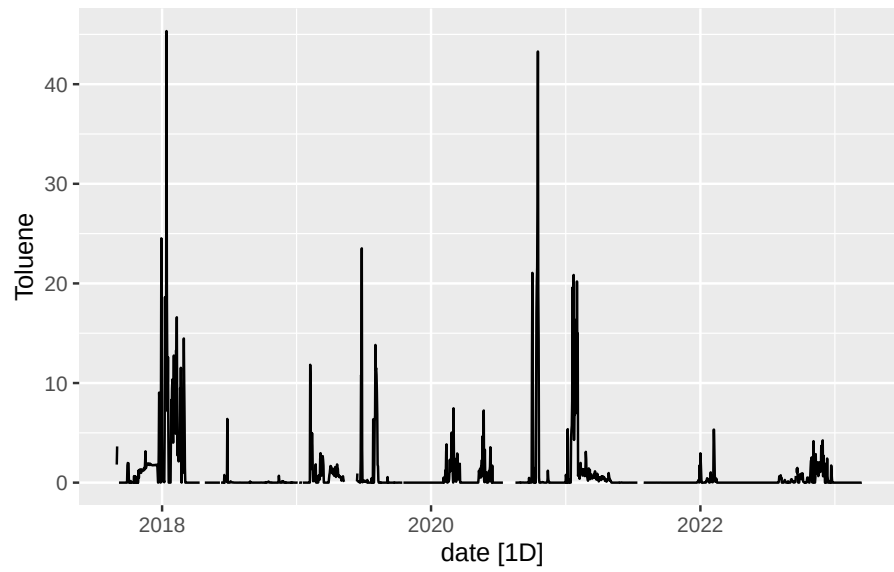
```
daily_df %>%  
  autoplot(CO)
```



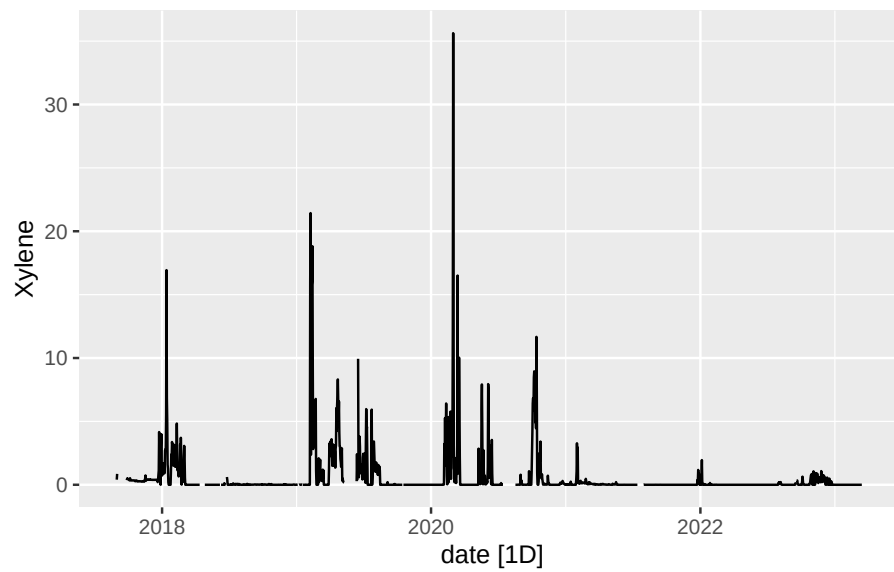
```
daily_df %>%  
  autoplot(Benzene)
```



```
daily_df %>%  
  autoplot(Toluene)
```

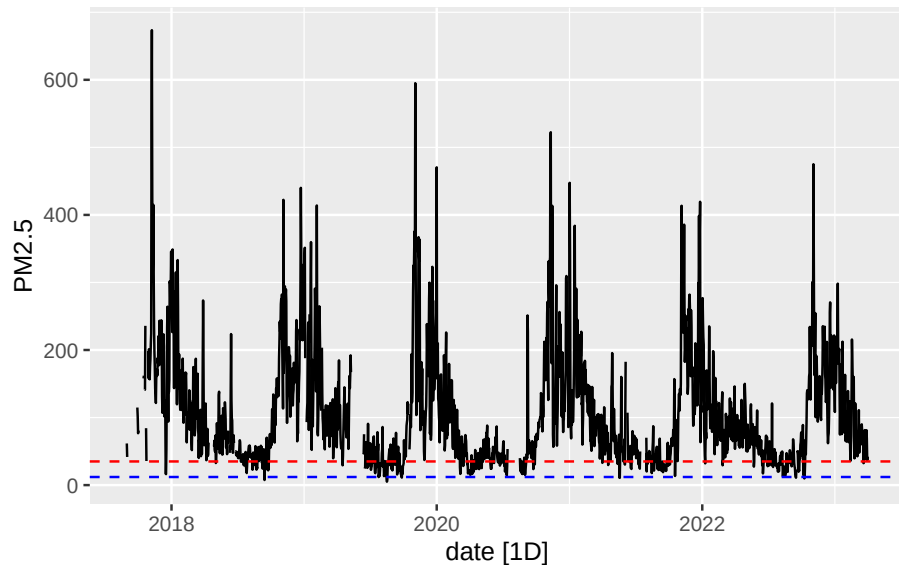


```
daily_df %>%  
  autoplot(Xylene)
```



Recomended Limit for PM2.5

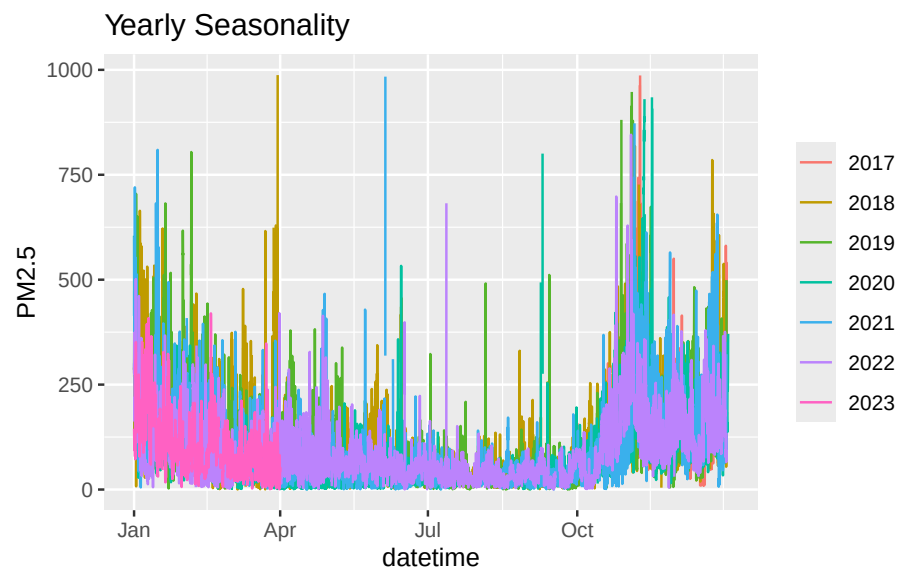
```
df %>%
  index_by(date = as.Date(datetime)) %>%
  summarise(PM2.5 = mean(PM2.5, na.rm = TRUE)) %>%
  autoplot(PM2.5) +
  geom_hline(yintercept = 35, color = "red", linetype = "dashed") +
  geom_hline(yintercept = 12, color = "blue", linetype = "dashed")
```



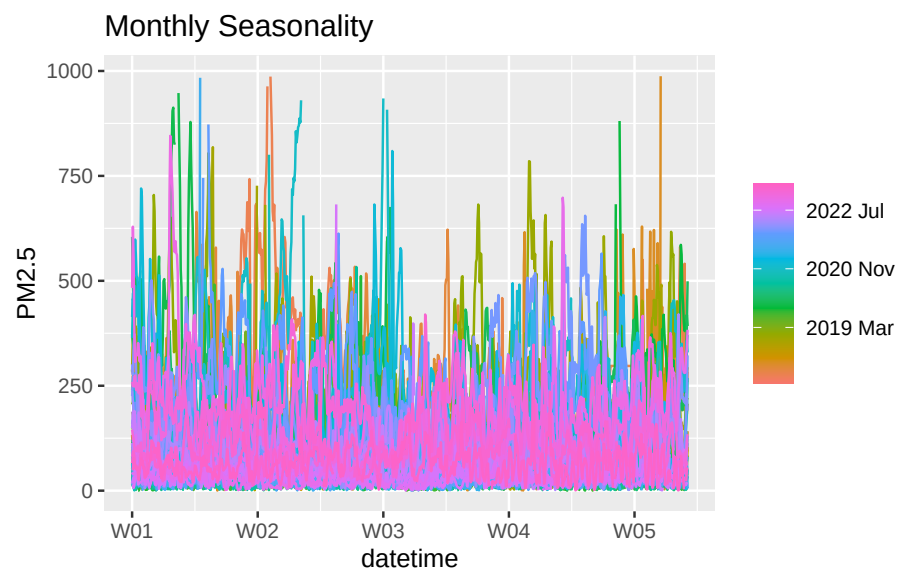
The World Health Organization recommends PM2.5 levels below $35\mu\text{g}/\text{m}^3$ for a daily average and below $12\mu\text{g}/\text{m}^3$ for a yearly average. The daily and yearly recommended levels are plotted in red and blue respectively. Only during the summer months are the PM2.5 levels below the WHO recommended daily limit.

Seasonality

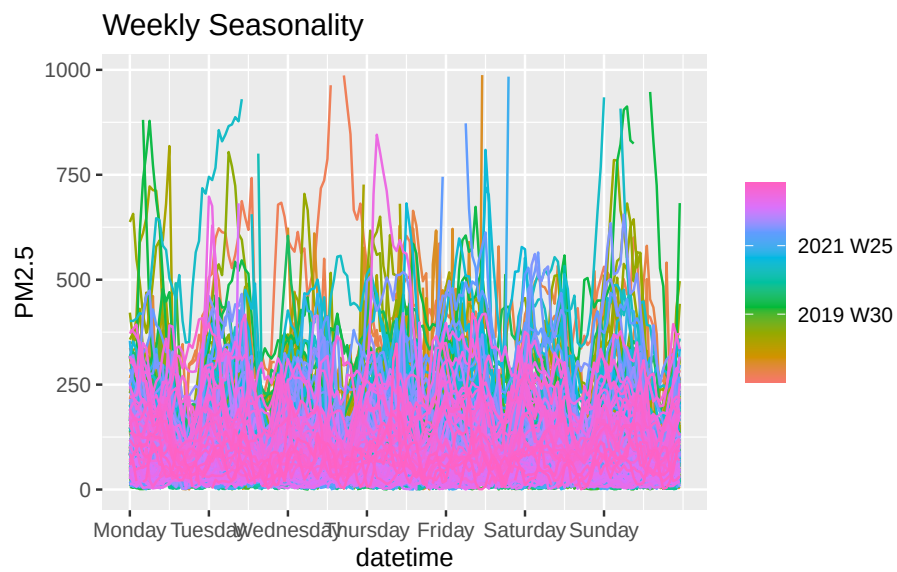
```
gg_season(df, PM2.5, period = "1 year") +
  labs(title = "Yearly Seasonality")
```



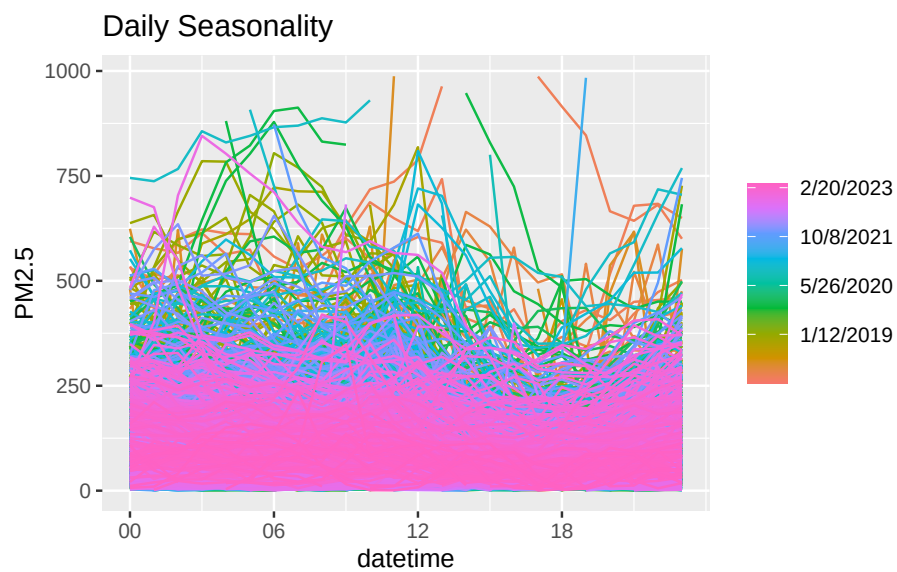
```
gg_season(df, PM2.5, period = "1 month") +
  labs(title = "Monthly Seasonality")
```



```
gg_season(df, PM2.5, period = "1 week") +
  labs(title = "Weekly Seasonality")
```



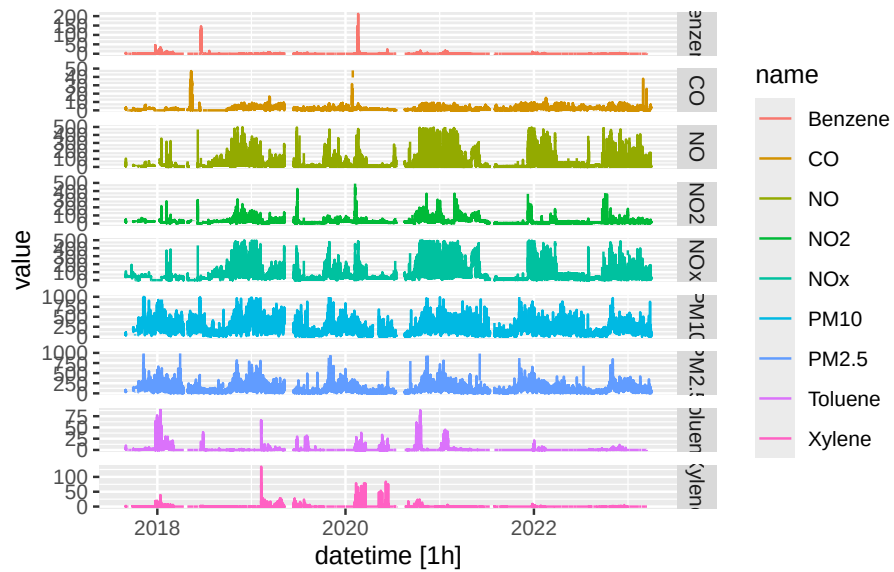
```
gg_season(df, PM2.5, period = "1 day") +
  labs(title = "Daily Seasonality")
```



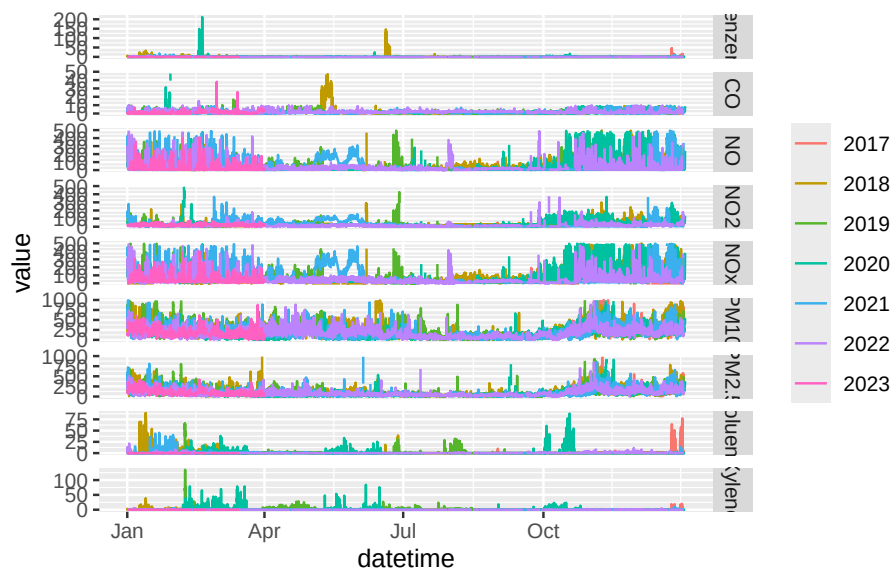
Plot Parameters Together

```
df_long <- df %>% pivot_longer(c(2:10))
```

```
df_long %>% autoplot(value) +  
  facet_grid(rows = vars(name), scales = "free_y")
```

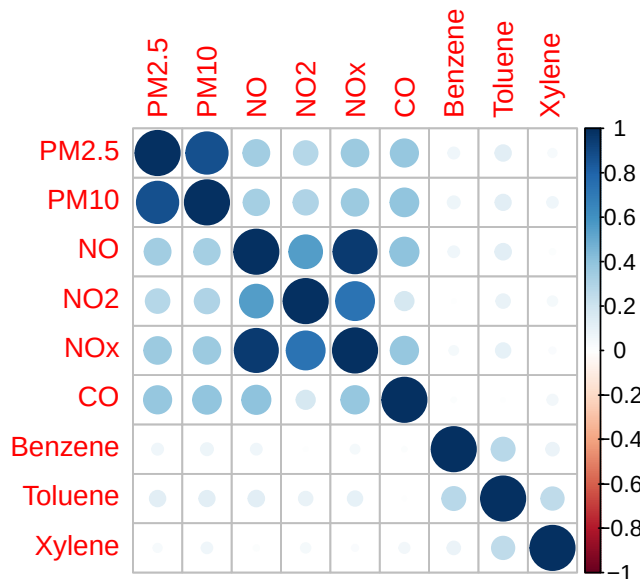


```
df_long %>% gg_season(value) +  
  facet_grid(rows = vars(name), scales = "free_y")
```



Parameter Correlations

```
M <- cor(x = df[, c(2:10)],
        y = df[, c(2:10)],
        use = "na.or.complete")
corrplot(M)
```



High concentrations of CO, NO, NO₂, NO_x, PM_{2.5}, and PM₁₀ generally occur at the same time. High concentrations of Benzene, Toluene, and Xylene generally occur at the same time.

PM_{2.5} as a Stationary Time Series

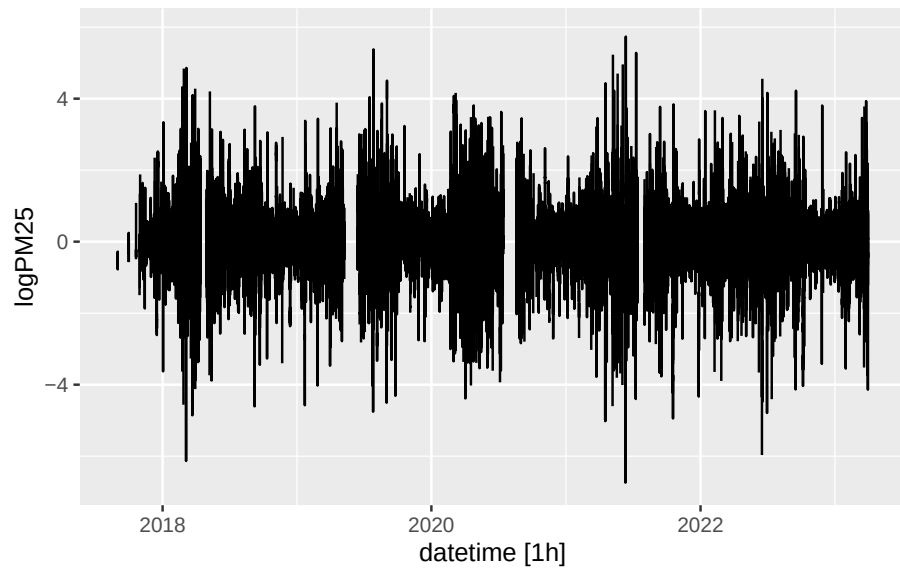
```
df$logPM25 <- log(df$PM2.5) %>%
  difference(24)
```

```
# Box-Cox Transformation
BoxCox.lambda(df$PM2.5)
```

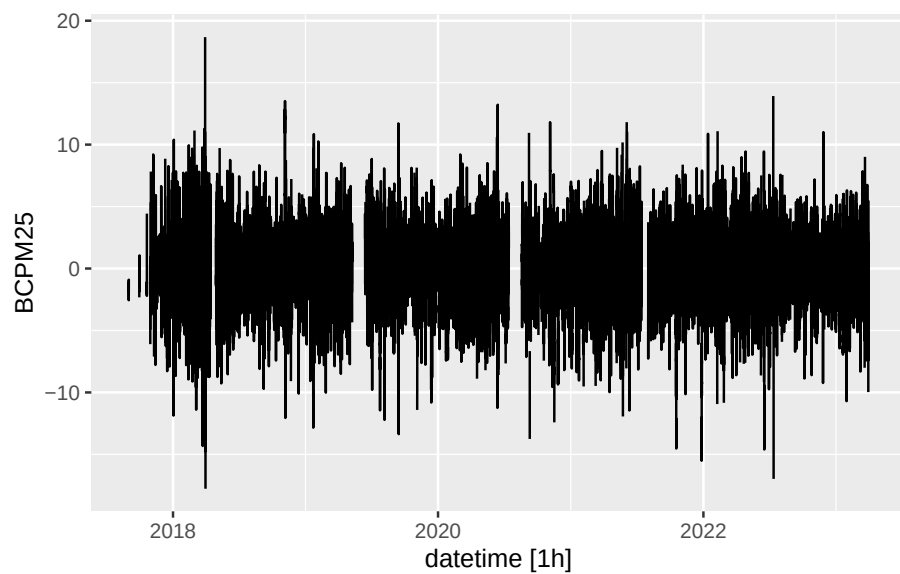
```
[1] 0.2959454
```

```
df$BCPM25 <- BoxCox(df$PM2.5, BoxCox.lambda(df$PM2.5)) %>%
  difference(24)
```

```
df %>% autoplot(logPM25)
```



```
df %>% autoplot(BCPM25)
```



Differencing PM2.5 centers the data, and using a log transform centers the data, giving a stationary time series. However, using a Box-Cox transform creates a more stationary time series than the log transform. This method of transformation is likely the better option if we were to move forward with a model choice that requires stationarity.

Check for Constant Variance

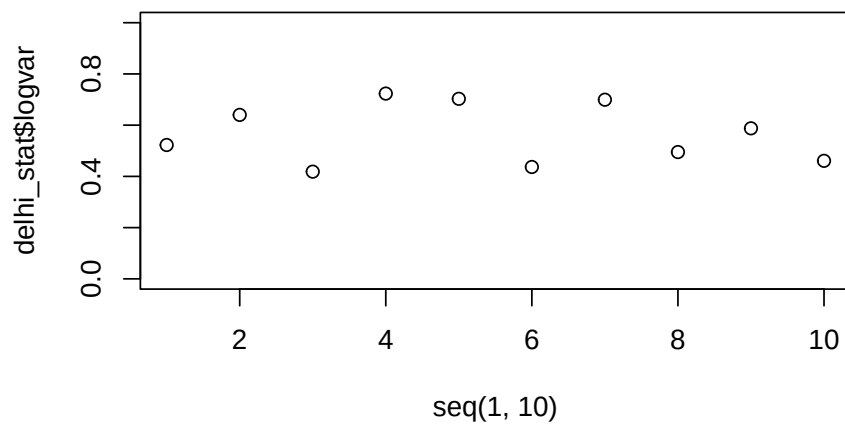
```
delhi_stat <- data.frame(logvar = numeric(10),
                        boxvar = numeric(10))
for (i in 1:10){
  delhi_stat[i,1] <- var(df$logPM25[
    floor(nrow(df)*0.1*(i-1)):floor(nrow(df)*0.1*i)
  ], na.rm = TRUE)

  delhi_stat[i,2] <- var(df$BCPM25[
    floor(nrow(df)*0.1*(i-1)):floor(nrow(df)*0.1*i)
  ], na.rm = TRUE)
}

delhi_stat
```

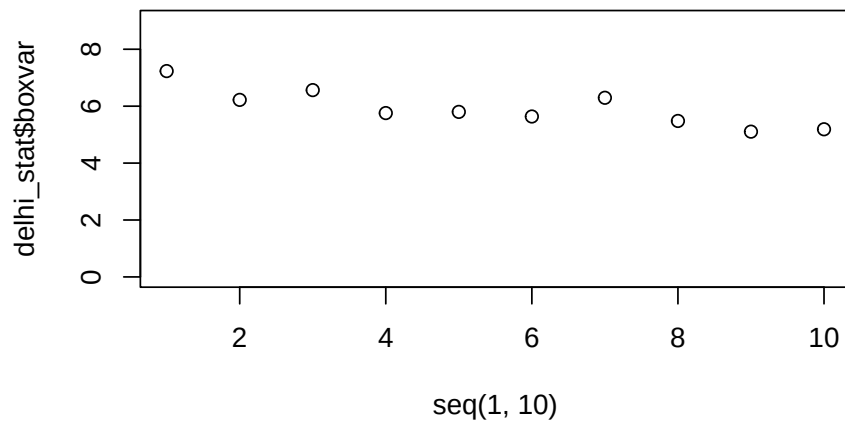
	logvar	boxvar
1	0.5225954	7.232752
2	0.6401363	6.220892
3	0.4184593	6.564328
4	0.7233174	5.757312
5	0.7029033	5.798939
6	0.4367190	5.636711
7	0.6993186	6.294659
8	0.4949893	5.480331
9	0.5876888	5.103950
10	0.4608093	5.187683

```
plot(seq(1,10),delhi_stat$logvar, ylim = c(0,1))
```



The variance changes without transformation.

```
plot(seq(1,10),delhi_stat$boxvar, ylim = c(0,9))
```



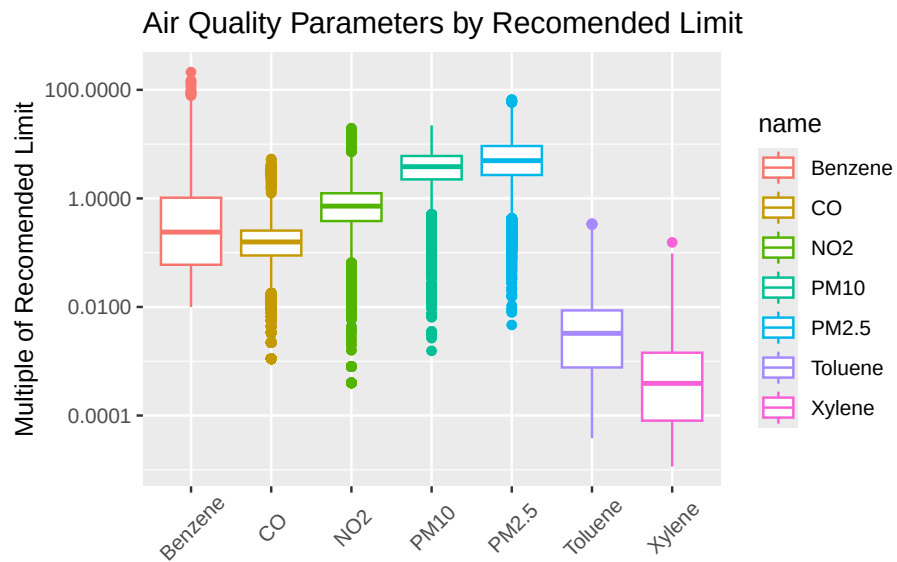
The Box-Cox transformation results in stable variances.

World Health Organization Recommended Limits:

CO: 9 ppm
PM₁₀: 15 µg/m³
PM_{2.5}: 45 µg/m³
NO₂: 25 µg/m³
Benzene: As low as possible (<5 µg/m³)
Toluene: 260 µg/m³
Xylene: 870 µg/m³

Scaled Parameters by Recommended Limits

```
df_scaled <- df %>%  
  select(-NOx, -NO) %>%  
  mutate(  
    PM2.5 = PM2.5 / 15,  
    PM10 = PM10 / 45,  
    CO = CO / 9,  
    NO2 = NO2 / 25,  
    Benzene = Benzene / 1,  
    Toluene = Toluene / 260,  
    Xylene = Xylene / 870  
  ) %>%  
  pivot_longer(c(2:8))  
  
ggplot(data = df_scaled) +  
  aes(x = name, y = value, color = name) +  
  geom_boxplot() +  
  scale_y_log10(labels = label_number()) +  
  theme(axis.text.x = element_text(angle = 45, vjust = 0.6)) +  
  labs(  
    x = NULL,  
    y = "Multiple of Recommended Limit",  
    title = "Air Quality Parameters by Recommended Limit"  
  )
```



Summary

Focus will be given to PM2.5 as it is a good indicator of overall air quality. It was found that PM2.5 is typically much above the recommended limits set by the World Health Organization. There is seasonality present, particularly at the yearly, daily, and hourly levels. Presence of PM2.5 is correlated with PM10, NO, NO2, NOx, and CO. While presence of Benzene, Toluene, and Xylene are not. It was found that a BoxCox transformation was able to transform the PM2.5 series to have constant variance. Future work will focus on creating various forecasting models for the PM2.5 series.